ImPRESS Statistical Analysis Report

Baseline Review for the Newly Diagnosed Glioblastoma Cohort

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Introduction

The ImPRESS study is a prospective investigation of losartan as an adjunctive therapy for patients with glioblastoma. According to Protocol Version 7.0 (17-Dec-2024), the trial collects comprehensive demographic, clinical, imaging, and quality-of-life information prior to treatment initiation. The statistical objectives for the interim analyses focus on characterising baseline status, neurological function, and tumour burden before randomisation to evaluate comparability across planned treatment arms.

This report summarises the currently available baseline data with emphasis on clinical status, imaging assessments, and prior treatments that may influence subsequent outcomes. The material is intended to provide early insight into data quality and cohort composition ahead of formal efficacy or safety analyses.

Randomisation Status

The present dataset is built on *sham randomisation* generated from the most recent export (see `_targets.R` workflow). No true treatment assignments have been unlocked. All findings in this document should therefore be interpreted as preliminary and descriptive only. Final efficacy comparisons will not be conducted until the database is locked and genuine randomisation codes are released.

Cohort Scope

The results herein are restricted to participants enrolled in the **newly diagnosed glioblastoma (AN) cohort**. Subjects with recurrent disease are excluded from all summaries, figures, and tables in this draft. A combined-cohort analysis will be delivered once recruitment and cleaning are complete for both strata.

Future revisions of this report will:

- incorporate the validated randomisation list once database lock is confirmed,
- integrate data from the recurrent glioblastoma cohort,
- extend the analyses to longitudinal efficacy and safety endpoints.

Baseline Characteristics

This section summarises baseline characteristics for the newly diagnosed glioblastoma cohort using the `adbl` ADaM dataset.

Demographics and Baseline characteristics

	all obs
Age (years)	
n	54
Mean (SD)	52.6 (10.2)
Median	55.0
Min - Max	26.0 - 69.0
Sex	
n	54
Female	29 (53.7%)
Male	25 (46.3%)
Cohort	
n	54
Newly diagnosed glioblastoma	54 (100%)
Country	
n	54
NOR	54 (100%)
Childbearing potential?	
n	29
No	13 (44.8%)
Yes	16 (55.2%)
Diagnosis	
n	54
Newly diagnosed glioblastoma	54 (100%)
Time from surgery to randomisation (days)	
n	54
Mean (SD)	27.4 (7.3)
Median	25.5
Min - Max	13.0 - 48.0
IDH1/2 mutation	
n	54
Negative	47 (87%)
Positive	7 (13%)
MGMT status	
n	54
Negative	30 (55.6%)

Demographics and Baseline characteristics

	all obs
Positive	24 (44.4%)
Pregnancy test result	
n	13
Negative	13 (100%)
ECOG Score	
n	54
ECOG Score0	40 (74.1%)
ECOG Score1	13 (24.1%)
ECOG Score2	1 (1.9%)
ECOG Score (numeric)	
n	54
Mean (SD)	0.3 (0.5)
Median	0.0
Min - Max	0.0 - 2.0
Karnofsky Performance Scale	
n	54
KPS 100	34 (63%)
KPS 60	1 (1.9%)
KPS 80	4 (7.4%)
KPS 90	15 (27.8%)
Karnofsky Performance Scale (numeric)	
n	54
Mean (SD)	95.0 (8.0)
Median	100.0
Min - Max	60.0 - 100.0
Total NANO score	
n	54
Mean (SD)	1.0 (1.3)
Median	0.0
Min - Max	0.0 - 4.0
EORTC QLQ-C30 score	
n	53
Mean (SD)	56.3 (8.6)
Median	54.0

Demographics and Baseline characteristics

	all obs
Min - Max	41.0 - 77.0
EORTC QLQ-BN20 score	
n	53
Mean (SD)	30.8 (6.6)
Median	30.0
Min - Max	20.0 - 49.0
Systolic blood pressure	
n	54
Mean (SD)	130.0 (15.4)
Median	128.5
Min - Max	106.0 - 180.0
Diastolic blood pressure	
n	54
Mean (SD)	81.9 (7.1)
Median	82.0
Min - Max	67.0 - 97.0
Pulse Rate	
n	54
Mean (SD)	80.7 (15.7)
Median	78.0
Min - Max	54.0 - 114.0
Weight (kg)	
n	54
Mean (SD)	78.8 (13.2)
Median	78.3
Min - Max	48.2 - 116.0
BMI	
n	51
Mean (SD)	25.0 (4.2)
Median	23.9
Min - Max	18.6 - 39.9
Height (cm)	
n	51
Mean (SD)	176.5 (8.8)

Demographics and Baseline characteristics

	all obs
Median	175.0
Min - Max	158.0 - 194.0
Has the patient received chemotherapy	
n	54
No	54 (100%)
Has the patient received immunotherapy	
n	54
No	54 (100%)
Has the patient received radiation therapy to the brain?	
n	54
No	54 (100%)
CNS surgery	
n	54
No	3 (5.6%)
Yes	51 (94.4%)
Steroids at time of MRI	
n	54
No	42 (77.8%)
Yes	12 (22.2%)
Mean relative cerebral blood flow (PTA-ROI)	
n	54
Mean (SD)	1.5 (0.4)
Median	1.4
Min - Max	0.7 - 2.4
Mean relative cerebral blood flow (ROI2)	
n	47
Mean (SD)	1.1 (0.5)
Median	1.0
Min - Max	0.2 - 2.2
Mean relative solid stress (PTA-ROI)	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1

Demographics and Baseline characteristics

	all obs
Mean relative solid stress (ROI2)	
n	32
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.5 - 1.4
Mean relative apparent diffusion coefficient	
n	52
Mean (SD)	1.4 (0.2)
Median	1.4
Min - Max	1.0 - 2.0
Mean relative fractional anisotropy	
n	52
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean relative restricted spectrum imaging	
n	50
Mean (SD)	0.5 (0.2)
Median	0.5
Min - Max	0.2 - 1.1
Mean DSC K2 leakage	
n	54
Mean (SD)	0.2 (1.6)
Median	0.4
Min - Max	-3.9 - 4.5
Mean relative cerebral blood	
n	54
Mean (SD)	1.7 (0.5)
Median	1.7
Min - Max	1.0 - 2.8
Mean relative transit time	
n	54
Mean (SD)	1.4 (0.3)
Median	1.4

Demographics and Baseline characteristics

	all obs
Min - Max	0.9 - 2.3
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	
n	54
Mean (SD)	6.3 (6.5)
Median	4.3
Min - Max	0.0 - 26.3
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	
n	54
Mean (SD)	13.5 (14.3)
Median	8.5
Min - Max	0.1 - 57.4
Mean tissue elasticity	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.2
Mean tissue relative viscosity	
n	41
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.0 - 1.1
Mean tissue stiffness	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean tissue viscosity	
n	41
Mean (SD)	0.7 (0.2)
Median	0.7
Min - Max	-0.0 - 1.1
Mean relative DSC extravasation	
n	28
Mean (SD)	1.2 (0.8)

Demographics and Baseline characteristics

	all obs
Median	1.0
Min - Max	0.7 - 5.1
Mean relative direction value	
n	28
Mean (SD)	0.4 (1.4)
Median	0.3
Min - Max	-2.7 - 5.4
Mean relative vessel caliber	
n	28
Mean (SD)	1.0 (0.3)
Median	1.0
Min - Max	0.5 - 1.9
Mean relative vessel size index	
n	28
Mean (SD)	1.4 (0.6)
Median	1.2
Min - Max	0.6 - 2.6
Mean relative vortex area	
n	28
Mean (SD)	1.0 (0.1)
Median	1.0
Min - Max	0.8 - 1.1